# Research note 04 - Local embeddings

Embeddings are generated on the user’s own server, while the plaintext is still in memory. Vectors never leave the trusted environment and are never written to Telegram. Only ciphertext is uploaded.

The query vector is encoded with the same model as the document text, so cosine similarity compares meaning rather than exact words.

# Compression policy

Plain text, code, CSV and JSON are compressed with zstd level 3.
JPEG, PNG, MP4 and archives are stored as-is: measurements show a saving of roughly one percent, which does not justify the processor time.

Office documents and PDFs are sampled (64 KiB) before the decision.

# Storage map

storage part    : 100 MiB, one Telegram message each
encryption block: 1 MiB, independent AES-256-GCM nonce and tag

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Seeking is mapped from a byte range to the covering blocks, so playback starts after the first block instead of the whole file.